

Histology · Board A · Slide images to function · sheet 1

Print on card stock. Cut on the card borders. Codes print on this side only, so the word cards give nothing away.

C01 Paste image C01	C02	C03
C04 Paste image C04	C05	C06 Paste image C06
C07 Paste image C07	C08	C09 Paste image C09
C10	C11	C12

Histology · Board A · Slide images to function · sheet 2

Print on card stock. Cut on the card borders. Codes print on this side only, so the word cards give nothing away.

C13 Paste image C13	C14 Paste image C14	C15 Paste image C15
C16	C17	C18
C19	C20	C21
C22	C23	C24

Histology · Board A · Slide images to function · sheet 3

Print on card stock. Cut on the card borders. Codes print on this side only, so the word cards give nothing away.

C25	C26	C27
C28	C29	C30
C31	C32	C33
C34	C35	C36

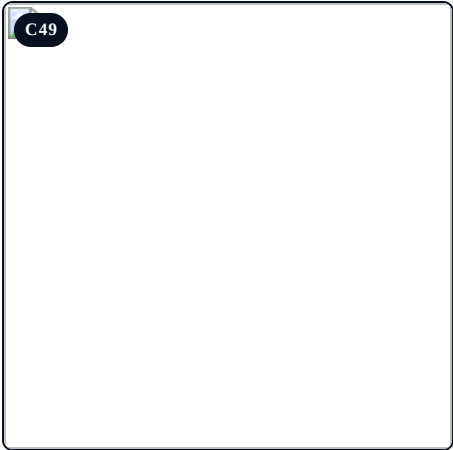
Histology · Board A · Slide images to function · sheet 4

Print on card stock. Cut on the card borders. Codes print on this side only, so the word cards give nothing away.

C37	C38	C39
C40	C41	C42
C43	C44	C45
C46	C47	C48

Histology · Board A · Slide images to function · sheet 5

Print on card stock. Cut on the card borders. Codes print on this side only, so the word cards give nothing away.



Histology · Board B · Function · sheet 1

Print on a second color of card stock so the two boards stay separate in the box.

Lets material pass quickly by diffusion and filtration across a very thin barrier.

Secretes and absorbs.

Absorbs nutrients and secretes mucus and digestive enzymes.

Moves material along the surface with beating cilia.

Secretes mucus and sweeps it back out of the airway with cilia.

Protects against abrasion and water loss with a tough surface of dead cells.

Protects surfaces that take heavy abrasion but have to stay moist.

Stretches flat as the organ fills and rounds back up as it empties.

Protects the walls of the larger gland ducts.

Wraps and cushions organs, holds tissue fluid, and houses defense cells.

Stores energy, insulates against heat loss, and pads structures.

Forms a soft internal framework that supports free blood cells.

Histology · Board B · Function · sheet 2

Print on a second color of card stock so the two boards stay separate in the box.

Resists a strong pull coming from one direction.

Resists pull arriving from many directions at once.

Lets a structure stretch and then recoil to its original shape.

Gives firm but flexible support and a smooth low friction surface.

Holds a definite shape while still bending easily.

Absorbs shock and resists heavy compression.

Supports and protects, gives muscles a lever to pull on, and stores calcium.

Gives strength at much lower weight and holds red bone marrow.

Carries gases, nutrients, and wastes around the body.

Moves the skeleton under voluntary control.

Pumps blood, involuntarily and rhythmically.

Squeezes hollow organs to move their contents along, involuntarily.

Histology · Board B · Function · sheet 3

Print on a second color of card stock so the two boards stay separate in the box.

Senses changes and carries electrical signals that control body activity.

The one cell layer that keeps dividing and replacing the whole epidermis.

Holds the epidermis together against pulling, using desmosomes between cells.

Where keratinocytes fill with granules, waterproof themselves, and start to die.

A clear extra layer that adds protection where friction is heaviest.

Forms the tough waterproof surface out of dead keratin filled cells.

Feeds the epidermis and carries touch receptors up into the dermal papillae.

Gives skin its strength and its resistance to stretching.

Releases watery sweat onto the skin surface to cool the body.

Releases a thicker sweat into a hair follicle, which skin bacteria then break down.

Releases oily sebum that softens the skin and the hair.

Produces a hair and anchors it in the skin.

Histology · Board B · Function · sheet 4

Print on a second color of card stock so the two boards stay separate in the box.

Detects light touch.

Detects deep pressure and vibration.

A single cell gland that makes and releases mucus.

Makes the fibers and the ground substance of connective tissue.

Stores fat in one large droplet that pushes the nucleus out to the edge.

Engulfs debris, dead cells, and pathogens.

Releases histamine and heparin to start an inflammatory response.

Maintains the cartilage matrix around it.

Maintains bone matrix and senses stress placed on the bone.

Makes melanin, which shields cell DNA from ultraviolet damage.

Makes keratin, the protein that toughens and waterproofs the epidermis.

Pairs with a nerve ending to act as a receptor for light touch.

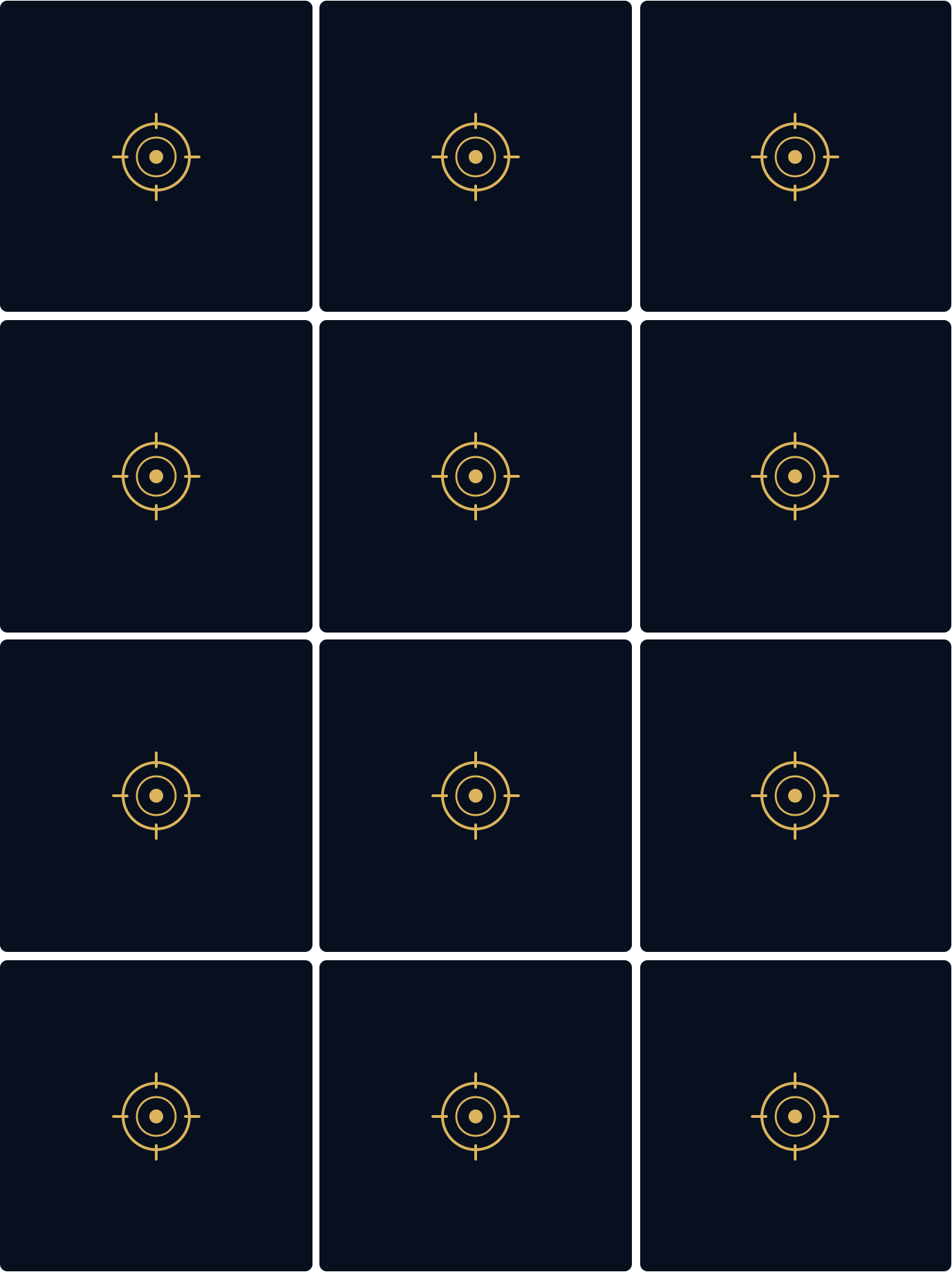
Histology · Board B · Function · sheet 5

Print on a second color of card stock so the two boards stay separate in the box.

Captures invaders in the skin and presents them to the immune system.

Card backs

Optional. Run the card sheets through again to back them, or print on pre-colored stock and skip this.



Answer key · Histology

Instructor copy. This one does not go in the card box.

CODE	TISSUE	WHAT IT DOES	WHERE YOU FIND IT
C01	Simple squamous epithelium	Lets material pass quickly by diffusion and filtration across a very thin barrier.	Air sacs of the lungs, the filtration membrane of the kidney, and the lining of blood vessels and serous membranes.
C02	Simple cuboidal epithelium	Secretes and absorbs.	Kidney tubules, thyroid follicles, the ducts of small glands, and the surface of the ovary.
C03	Simple columnar epithelium	Absorbs nutrients and secretes mucus and digestive enzymes.	Lining of the stomach, small intestine, and large intestine.
C04	Ciliated simple columnar epithelium	Moves material along the surface with beating cilia.	Lining of the uterine tubes, the uterus, and the smaller bronchi.
C05	Pseudostratified ciliated columnar epithelium	Secretes mucus and sweeps it back out of the airway with cilia.	Lining of the trachea and most of the upper respiratory tract.
C06	Keratinized stratified squamous epithelium	Protects against abrasion and water loss with a tough surface of dead cells.	Epidermis of the skin.
C07	Nonkeratinized stratified squamous epithelium	Protects surfaces that take heavy abrasion but have to stay moist.	Lining of the mouth, esophagus, and vagina.
C08	Transitional epithelium	Stretches flat as the organ fills and rounds back up as it empties.	Lining of the urinary bladder, the ureters, and part of the urethra.
C09	Stratified cuboidal epithelium	Protects the walls of the larger gland ducts.	Ducts of sweat glands, mammary glands, and salivary glands.
C10	Areolar connective tissue	Wraps and cushions organs, holds tissue fluid, and houses defense cells.	Under every epithelium as the lamina propria, in the papillary dermis, and packaging around organs.
C11	Adipose tissue	Stores energy, insulates against heat loss, and pads structures.	Under the skin, around the kidneys and eyes, inside the abdomen, and in the breasts.
C12	Reticular connective tissue	Forms a soft internal framework that supports free blood cells.	Lymph nodes, spleen, and red bone marrow.
C13	Dense regular connective tissue	Resists a strong pull coming from one direction.	Tendons, ligaments, and aponeuroses.
C14	Dense irregular connective tissue	Resists pull arriving from many directions at once.	Reticular layer of the dermis, joint capsules, and the fibrous capsules around organs.
C15	Elastic connective tissue	Lets a structure stretch and then recoil to its original shape.	Walls of the large arteries and the ligamenta flava between vertebrae.
C16	Hyaline cartilage	Gives firm but flexible support and a smooth low friction surface.	Ends of long bones at joints, costal cartilages, the nose, the rings of the trachea, and the larynx.
C17	Elastic cartilage	Holds a definite shape while still bending easily.	External ear and the epiglottis.

CODE	TISSUE	WHAT IT DOES	WHERE YOU FIND IT
C18	Fibrocartilage	Absorbs shock and resists heavy compression.	Intervertebral discs, the pubic symphysis, and the menisci of the knee.
C19	Compact bone	Supports and protects, gives muscles a lever to pull on, and stores calcium.	Shaft of a long bone and the outer shell of every bone.
C20	Spongy bone	Gives strength at much lower weight and holds red bone marrow.	Ends of long bones and the inside of flat bones such as the sternum and hip bone.
C21	Blood	Carries gases, nutrients, and wastes around the body.	Inside the heart chambers and the blood vessels.
C22	Skeletal muscle tissue	Moves the skeleton under voluntary control.	Attached to bones, plus a few muscles attached to the skin of the face.
C23	Cardiac muscle tissue	Pumps blood, involuntarily and rhythmically.	Wall of the heart.
C24	Smooth muscle tissue	Squeezes hollow organs to move their contents along, involuntarily.	Walls of the digestive tract, blood vessels, airways, bladder, and uterus.
C25	Nervous tissue	Senses changes and carries electrical signals that control body activity.	Brain, spinal cord, and nerves.
C26	Stratum basale	The one cell layer that keeps dividing and replacing the whole epidermis.	Deepest layer of the epidermis, sitting on the basement membrane.
C27	Stratum spinosum	Holds the epidermis together against pulling, using desmosomes between cells.	Several cell layers thick, directly above the stratum basale.
C28	Stratum granulosum	Where keratinocytes fill with granules, waterproof themselves, and start to die.	Middle of the epidermis, above the stratum spinosum.
C29	Stratum lucidum	A clear extra layer that adds protection where friction is heaviest.	Thick skin only, on the palms and the soles.
C30	Stratum corneum	Forms the tough waterproof surface out of dead keratin filled cells.	Outermost layer of the epidermis.
C31	Papillary layer of the dermis	Feeds the epidermis and carries touch receptors up into the dermal papillae.	Upper part of the dermis, just under the epidermis.
C32	Reticular layer of the dermis	Gives skin its strength and its resistance to stretching.	Deep part of the dermis, making up most of its thickness.
C33	Eccrine sweat gland	Releases watery sweat onto the skin surface to cool the body.	Coiled in the dermis, most dense on the palms, soles, and forehead.
C34	Apocrine sweat gland	Releases a thicker sweat into a hair follicle, which skin bacteria then break down.	Dermis of the axilla and the anogenital region.
C35	Sebaceous gland	Releases oily sebum that softens the skin and the hair.	Dermis, opening into hair follicles, absent from the palms and soles.

CODE	TISSUE	WHAT IT DOES	WHERE YOU FIND IT
C36	Hair follicle	Produces a hair and anchors it in the skin.	Runs from the epidermis down into the dermis and hypodermis.
C37	Tactile corpuscle	Detects light touch.	Dermal papillae of hairless skin such as fingertips, lips, and nipples.
C38	Lamellar corpuscle	Detects deep pressure and vibration.	Deep dermis and hypodermis.
C39	Goblet cell	A single cell gland that makes and releases mucus.	Scattered through the epithelium of the intestines and the respiratory tract.
C40	Fibroblast	Makes the fibers and the ground substance of connective tissue.	Connective tissue proper, both loose and dense.
C41	Adipocyte	Stores fat in one large droplet that pushes the nucleus out to the edge.	Adipose tissue.
C42	Macrophage	Engulfs debris, dead cells, and pathogens.	Areolar tissue and the lymphoid organs.
C43	Mast cell	Releases histamine and heparin to start an inflammatory response.	Areolar tissue, clustered along the small blood vessels.
C44	Chondrocyte	Maintains the cartilage matrix around it.	Inside a lacuna within cartilage.
C45	Osteocyte	Maintains bone matrix and senses stress placed on the bone.	Inside a lacuna within compact and spongy bone.
C46	Melanocyte	Makes melanin, which shields cell DNA from ultraviolet damage.	Stratum basale of the epidermis.
C47	Keratinocyte	Makes keratin, the protein that toughens and waterproofs the epidermis.	Every layer of the epidermis.
C48	Tactile epithelial cell	Pairs with a nerve ending to act as a receptor for light touch.	Stratum basale of the epidermis.
C49	Dendritic cell of the epidermis	Captures invaders in the skin and presents them to the immune system.	Stratum spinosum of the epidermis.

Table rules and team score sheet

How the two boards work

1. Lay Board A cards face down on the left of the table and Board B cards face down on the right. Keep the two piles apart.
2. On your turn, turn over one card from the left and one card from the right.
3. If they belong together, say out loud why before you keep them. Keep the pair, score a point, and go again.
4. If they do not belong together, turn both back face down in the same spots and play passes to the next team.
5. When every pair is gone, the team holding the most pairs wins the round.

Streak scoring, optional

1. First match in a turn is worth 1 point, the second in a row is worth 2, the third in a row is worth 3, and so on.
2. A miss resets the streak and passes the turn.

Rule that makes it teach

1. A pair only counts if the team says the reason out loud. Another team may challenge it, and the instructor settles it from the answer key.

TEAM	ROUND 1	ROUND 2	ROUND 3	ROUND 4	TOTAL